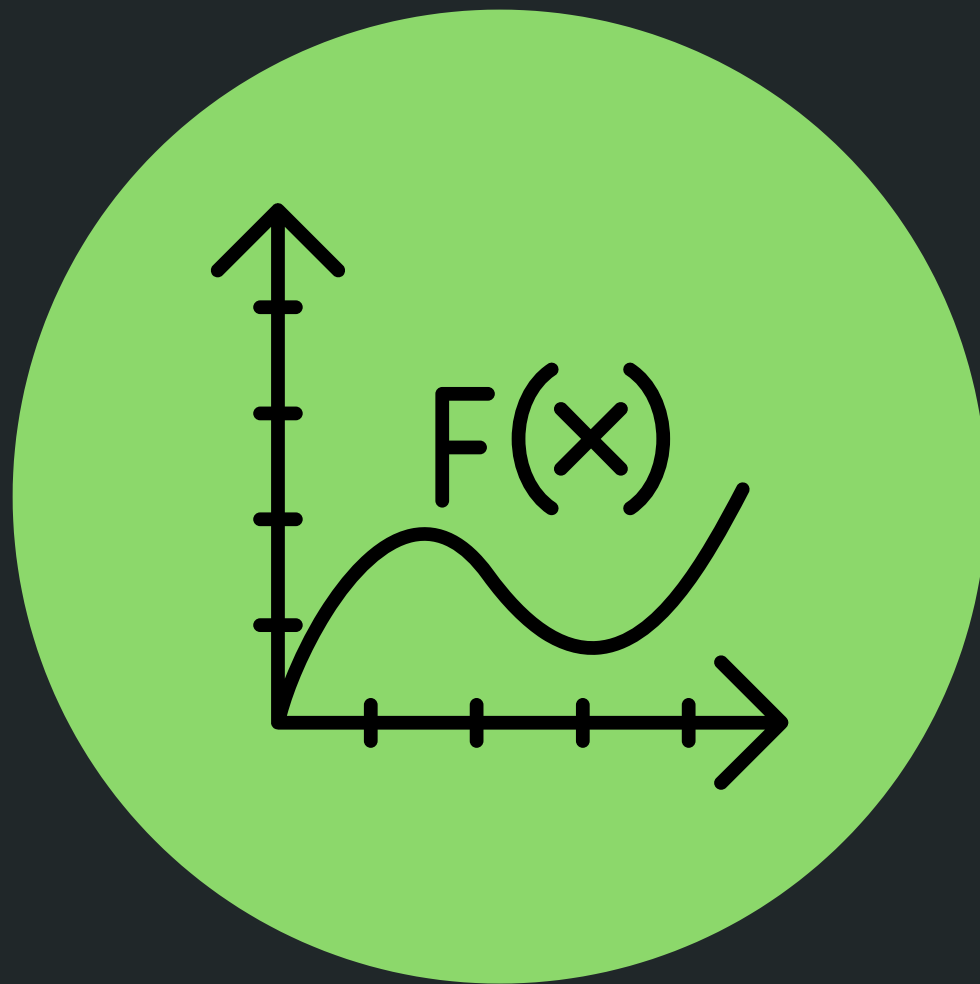


6.9. Calculus Basics



$$\frac{dy}{dx}$$

Why Calculus Matters in ML?



About Calculus

Calculus explains how things change. It helps measure rates, slopes, and movement in functions.

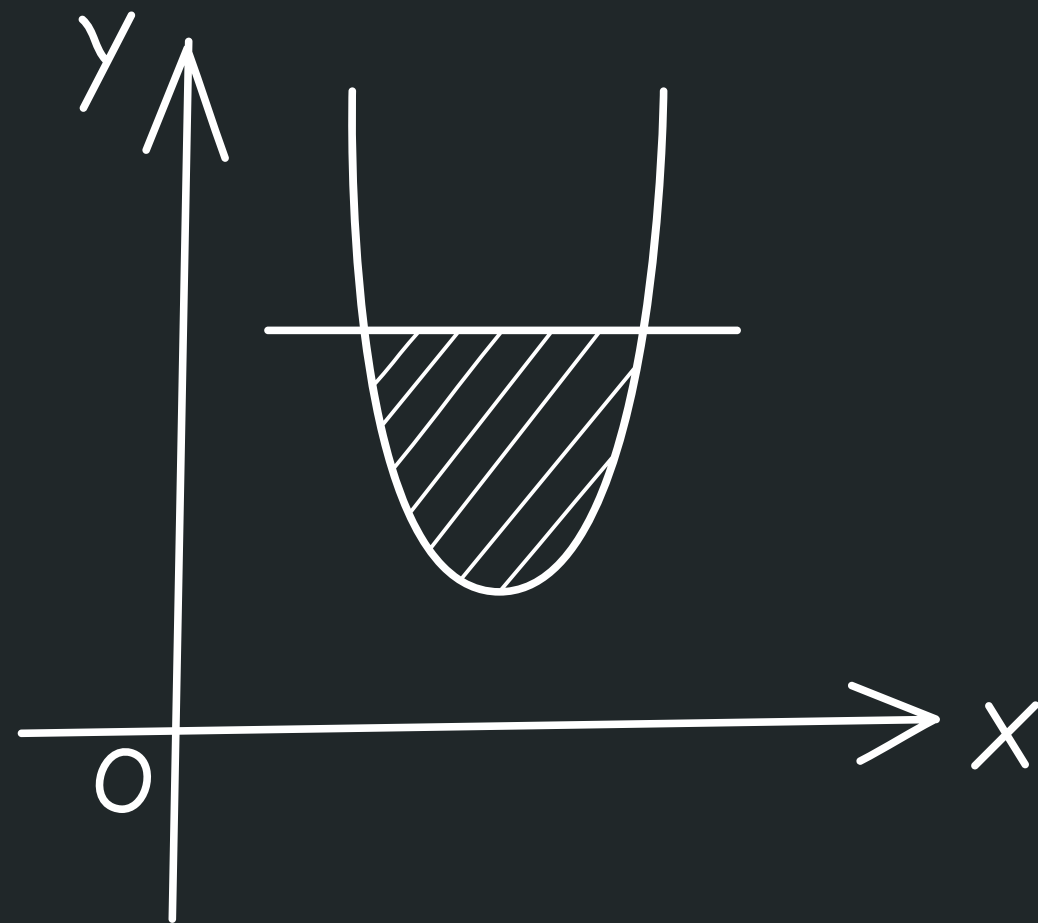
Purpose in ML

- Understand how predictions react to small changes
- Guide weight updates during training
- Improve model accuracy by following the right direction

Example

How does loss change when a weight increases slightly?

Functions in ML



What is a Function

A function maps input to output (e.g., $x \rightarrow y$).
Used in ML for predictions.

Purpose in ML

Models behave like functions where small input changes produce new predictions.

Example

Salary prediction based on years of experience.

Understanding Change



Change in Input

Small change in x



Change in Output

Resulting change in y

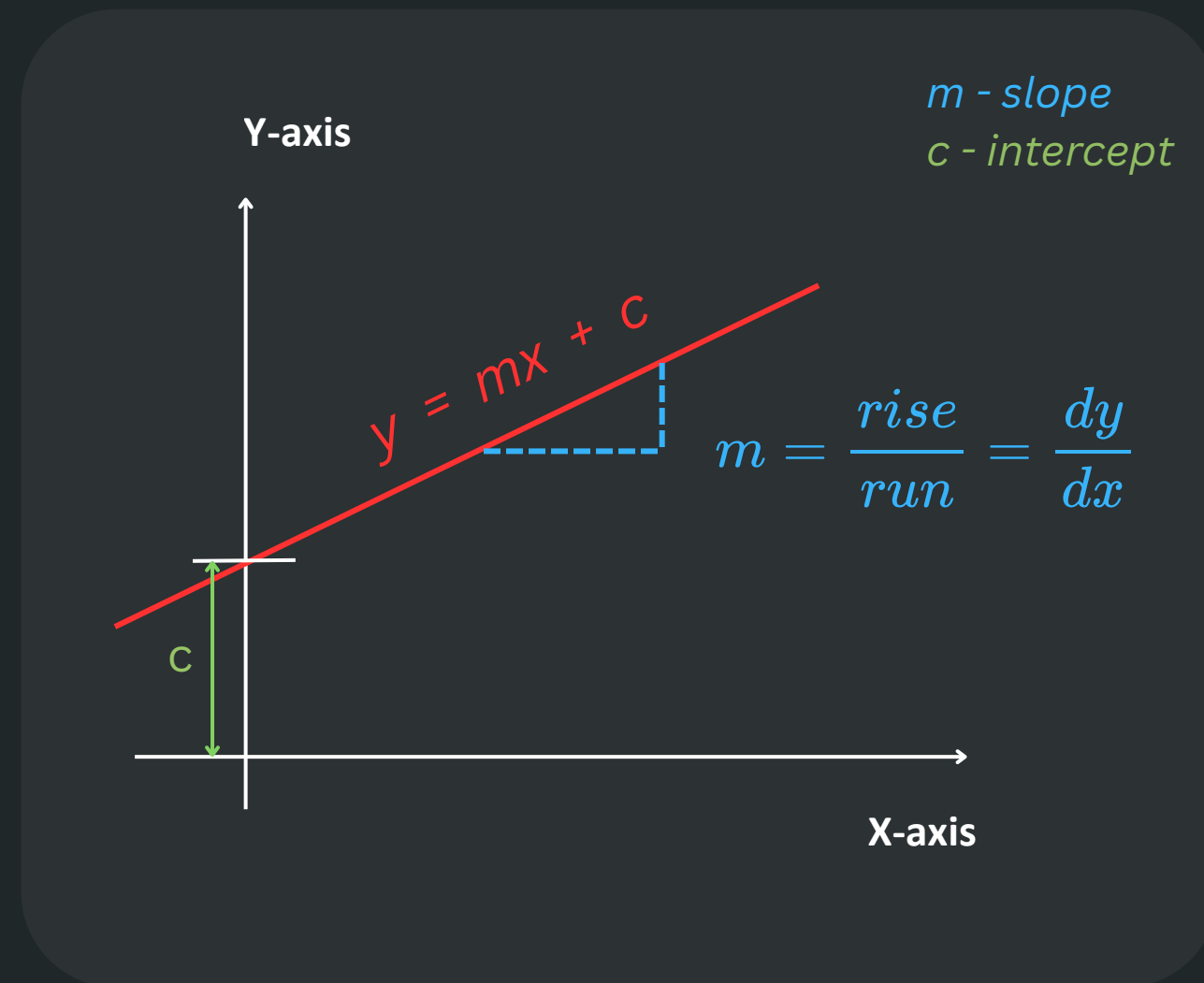


Rate of Change

How fast y changes

Use in ML: Helps measure sensitivity of model output.

Slope of a Line



Slope

Slope tells how steep a line is.

Slope = rise over run

(rise is change in y , run is change in x).

How to Read It

- Upward line $\rightarrow$ positive slope
- Downward line $\rightarrow$ negative slope
- Steeper line $\rightarrow$ bigger slope

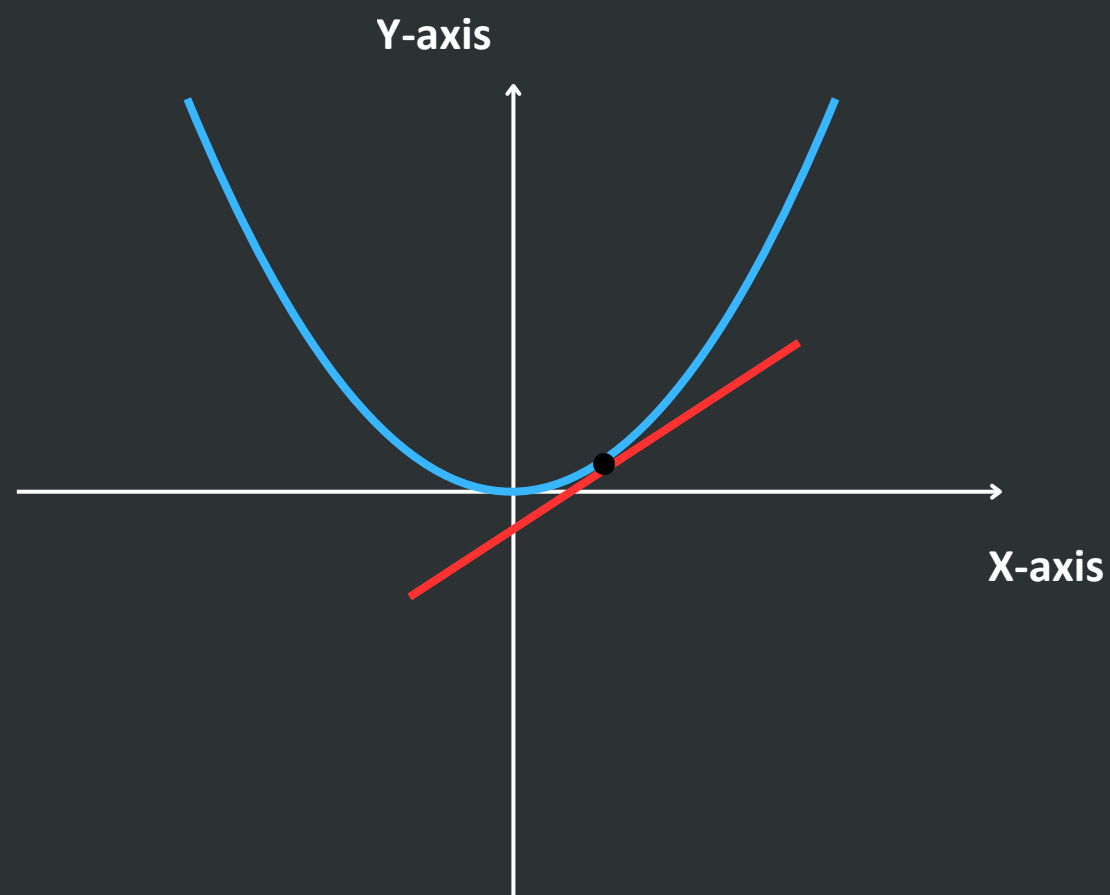
Importance in ML

Slope shows how fast the output changes when the input changes.

Example

A line with slope 2 rises twice as fast as x increases.

Slope on a Curve



Tangent Line

Slope at a point

Shows how steep the curve is right at that position.

Different Slope

Changes along the curve

Steep in some areas, flat in others because the curve bends.

Rate of Change (dy/dx)

How fast the curve changes

Tells how quickly the output changes for a small change in input.

Use in ML

Guides weight updates

Optimization - The slope tells the model which direction improves the loss.

Gradient (Multi Dimensional Slope)



What It Is

Slope in many dimensions.
Each weight in a model has its own slope.
This full set of slopes is the gradient.

Importance in ML

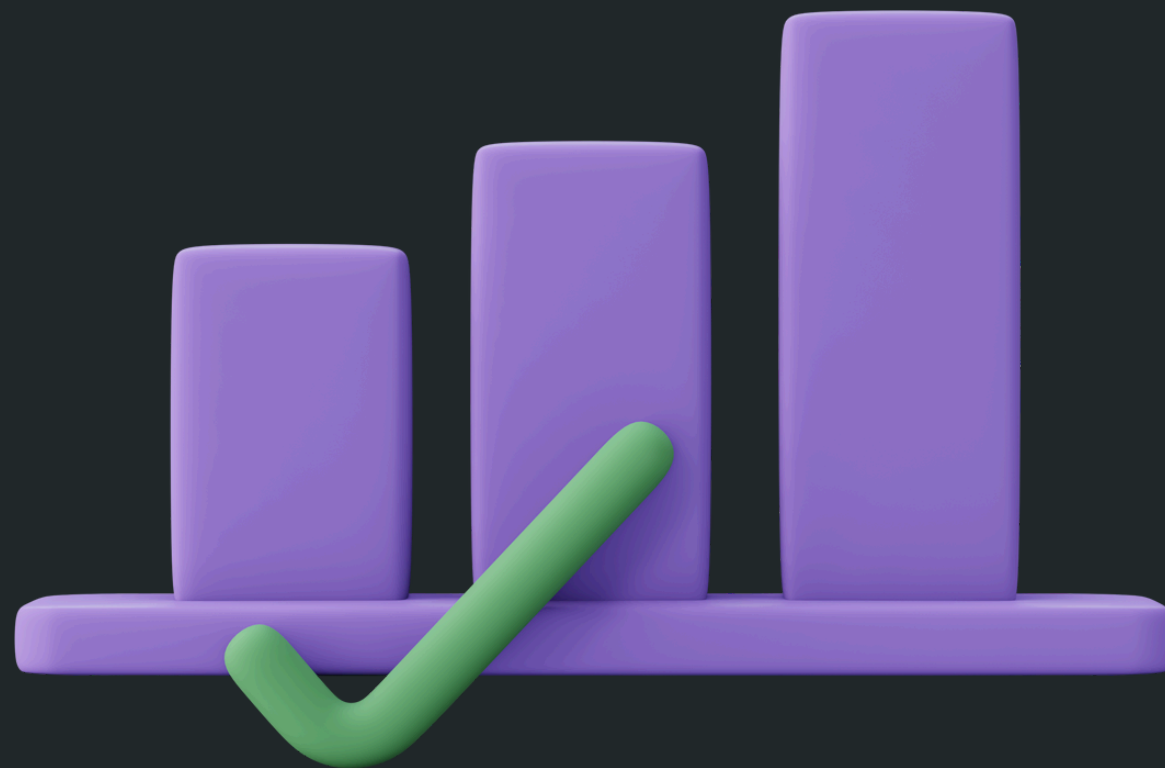
- Shows fastest direction of change
- Used to reduce loss
- Helps tune all parameters together

Why It Matters

Gradient tells the model how to update weights

- Big gradient → adjust weight strongly
- Small gradient → adjust weight gently
- Zero gradient → model has stopped improving

Loss Function (What We Minimize)



What It Is

Measure of error

Shows how far predictions are from the actual values.

Example

U shaped curve (MSE)

Lower value means better performance.

Sample Formula

Mean Squared Error (MSE)

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

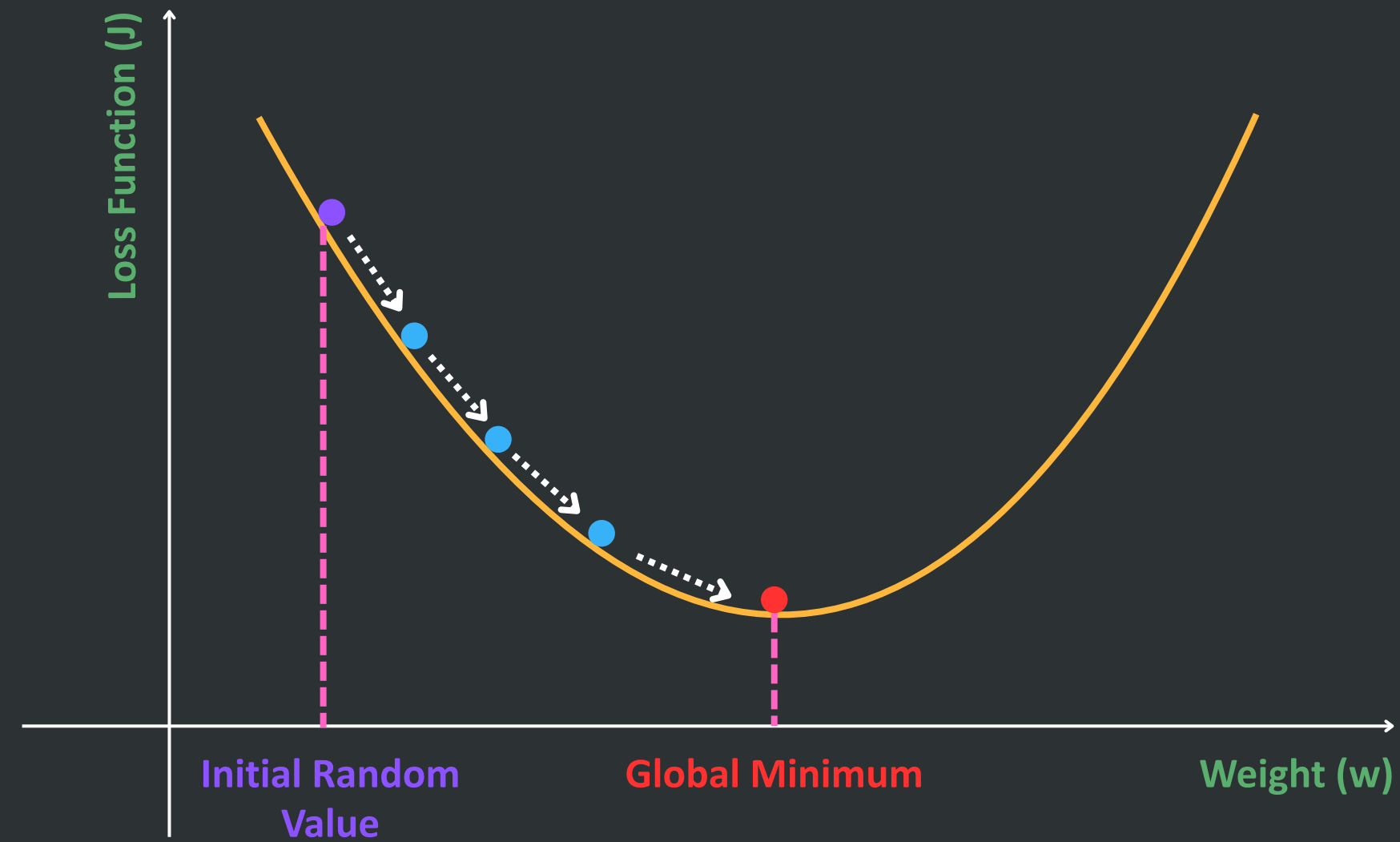
Why It Matters

Goal is to reduce loss

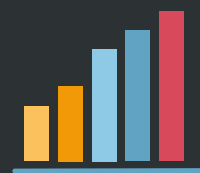
Lower loss means more accurate predictions.

Gradient Descent

Gradient Descent is a method that helps a model learn by moving its weights step by step in the direction that reduces the loss. It follows the slope of the loss curve and keeps moving downhill until it reaches the lowest point.



How Models Learn



Predict

Model makes an output.



Calculate Loss

Compare prediction vs actual



Update Weights

Use gradient descent to reduce the loss.

Why Calculus Matters



Training

Use derivatives to understand change



Optimisers

Use gradients to update weights



Loss Reduction

Follow slopes to minimise error